Revealing the Cosmic Dark Ages from lunar orbit with CosmoCube

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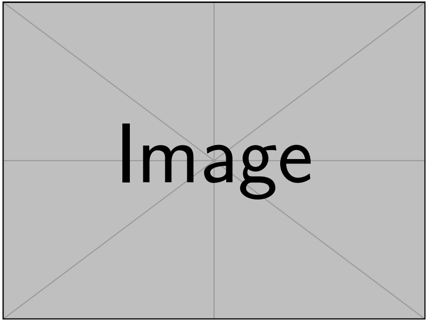
The Cosmic Dark Ages

The Sky Averaged 21-cm Signal

- Arises from the spin flip transition in neutral hydrogen
- Statistical temperature which quantifies the number of atoms in each state
- Measure this temperature relative to the CMB over cosmological time
- Transition driven by CMB photons, collisions between atoms and eventually astrophysics
- The dark ages is purely cosmology



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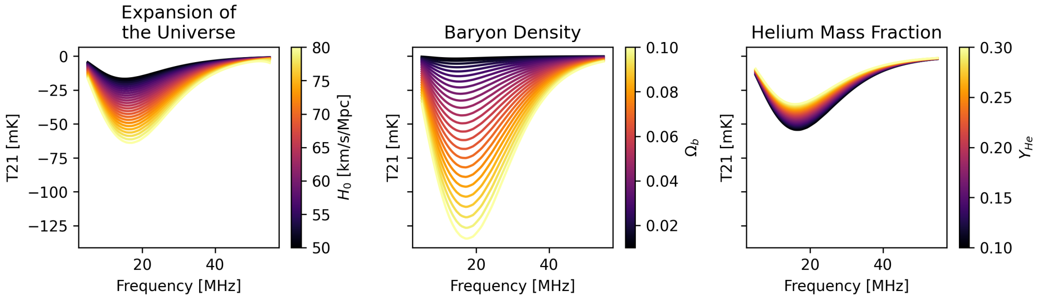
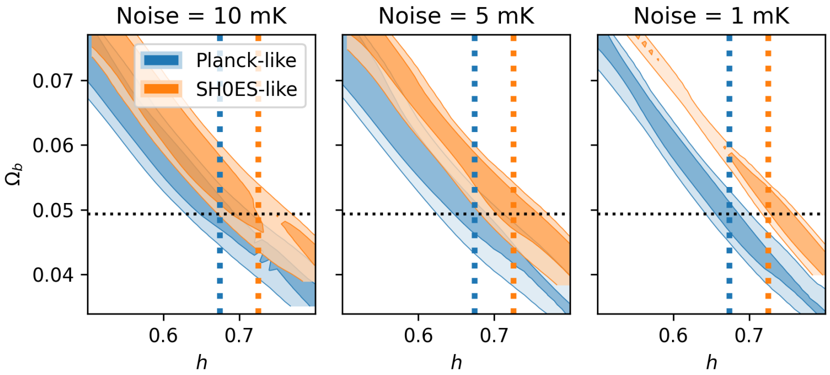
Λ-CDM Cosmology

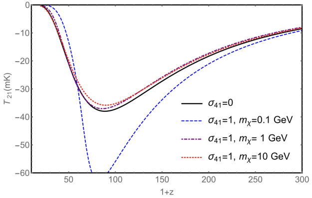


Using `hyprfine` code to exploe how the 21-cm signal varies around a fiducial Planck Cosmology. See [`https://github.com/htjb/hyprfine`](https://github.com/htjb/hyprfine).

Λ-CDM Cosmology

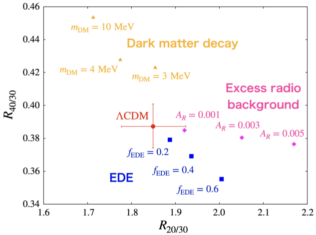


Beyond Λ-CDM



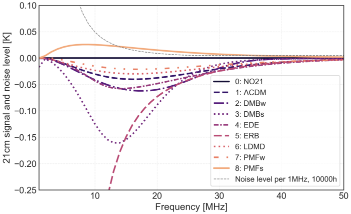
Munoz et al. PRD 2015

- DM-Baryon Interactions



Okamatsu et al. PRL 2024

- DM decay
- EDE
- Excess Radio Background



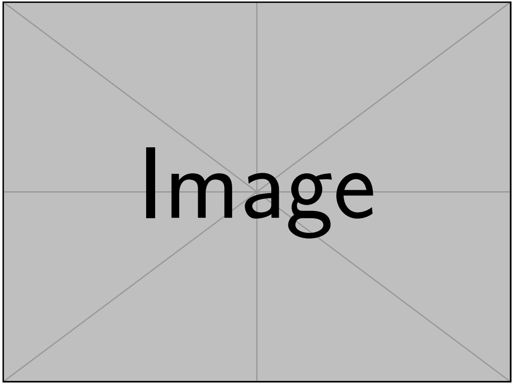
Yoshiura et al. 2026

- Primordial Magnetic Fields

Lunar orbit, CosmoCube and the race back to the Moon

Why go to the moon?

- Bullet pointing to the figure on the right
- Another bullet point
- And another one
- Sub-item example:
    - Sub-point one
    - Sub-point two



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Radio Astronomy on the Moon?

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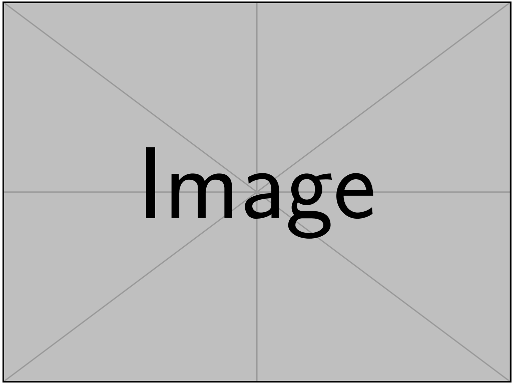

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CosmoCube

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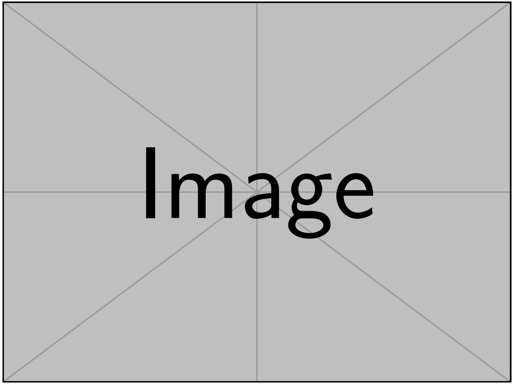

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Progress with CosmoCube

The Platform

- Bullet pointing to the figure on the right
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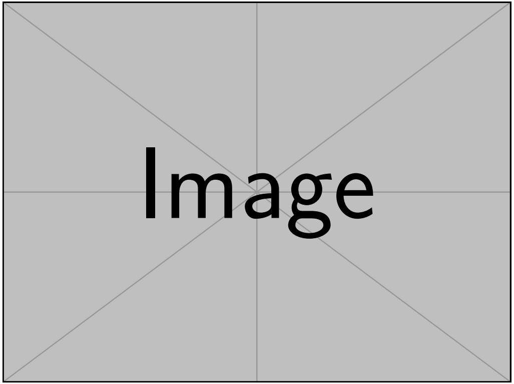

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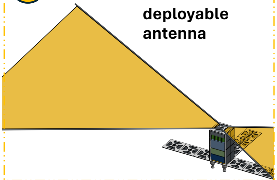
The Payload

1

Wideband deployable antenna



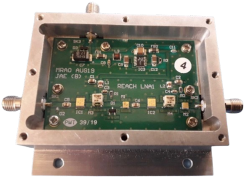
3

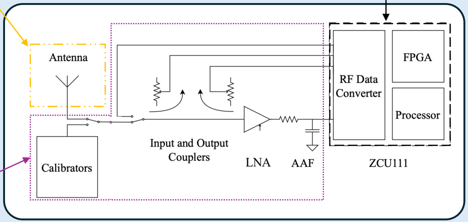
Fully integrated RFSoC receiver

- ✓ ADC
- ✓ Spectrometer
- ✓ VNA for calibration
- ✓ Basic onboard processor

2

Low Noise Amplifier and calibrators





CosmoCube payload diagram

Main payload specifications

6-m wideband dipole
Single polarization
Deployable structure
Low achromatic losses

"Dicke" switch calibrator
15 calibrators
Switches – IL < 0.5 dB
Switches – Iso > 60 dB
Low Noise Amplifier with
S_{11}<-30 dB

ADC + Spectrometer
8 – 46 [100] MHz
14 bits ADC
10 KHz narrow channels
100 KHz wide channels
SFDR > 70dBc
VNA
1 channel VNA function
$|\Delta S_{11}|$<0.05dB

The Journey to Launch

- Bullet pointing to the figure on the right
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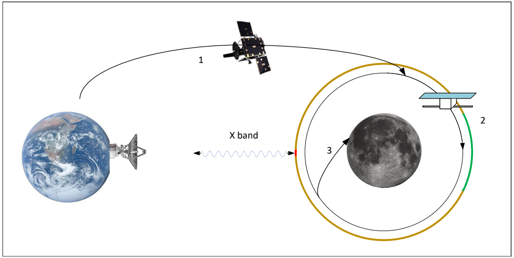

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Summary

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- Highlighted text in navy blue
- Back to normal text